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UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-33 20kV FlourMill SCHEMATIC DIAGRAM DRC BUS PT H05																							
				Date Prep.		15.11.2018		PROJECT TITLE				ABB		EPMV		COVER SHEET		Scale				= H05	
				Prepared		YR		MV Switchgear for MNA Serang Plant				PT. ABB Sakti Industri		MVD-33 20kV FlourMill SCHEMATIC DIAGRAM DRC BUS PT		Customer		PT WILMAR NABATI INDONESIA		& 308			
00		Issued for Approval		22.02.19		YR										Checked		AW		Drawing Nbr.		18.042-WNI-308.01	
Ind.		Revision		Date		Name		Approved		AI		Based on: BE318043				Customer Doc. Nbr.				Cont.		16	
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SHEET INDEX

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==MVD-33=H05&308/6		REFERENCE DESIGNATIONS		00		
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	Index of Sheet MVD-33 20kV FlourMill SCHEMATIC DIAGRAM DRC BUS PT	Scale		= H05	
				Prepared	YR						Customer	PT WILMAR NABATI INDONESIA & 308		
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A	REFERENCE DESIGNATION OF OBJECTS IN ELECTRICAL DOCUMENTS														A
	(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)														
B	DESIGNATION	DESCRIPTION													B
	-AA	MULTIFUNCTION UNIT (CENTRAL UNIT)													
	-BAD1,-BAD4	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L1													
	-BAD2,-BAD5	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L2													
	-BAD3,-BAD6	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L3													
	-BAR	VOLTAGE PROTECTION RELAY													
	-BAS1	VOLTAGE SENSOR LOCATED ON PHASE L1													
	-BAS2	VOLTAGE SENSOR LOCATED ON PHASE L2													
	-BAS3	VOLTAGE SENSOR LOCATED ON PHASE L3													
	-BAT1	VOLTAGE TRANSFORMER LOCATED ON PHASE L1													
C	-BAT2	VOLTAGE TRANSFORMER LOCATED ON PHASE L2													C
	-BAT3	VOLTAGE TRANSFORMER LOCATED ON PHASE L3													
	-BAT4	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L1													
	-BAT5	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L2													
	-BAT6	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L3													
	-BCD	DIFFERENTIAL PROTECTIVE RELAY													
	-BCF	FEEDER PROTECTION RELAY													
	-BCG	GENERATOR PROTECTION RELAY													
	-BCM	MOTOR PROTECTION RELAY													
	-BCN	NEUTRAL (RESIDUAL) CURRENT TRANSFORMER													
D	-BCP	TRANSFORMER PROTECTION RELAY													D
	-BCR	CURRENT PROTECTION RELAY													
	-BCS1	CURRENT SENSOR LOCATED ON PHASE L1													
	-BCS2	CURRENT SENSOR LOCATED ON PHASE L2													
	-BCS3	CURRENT SENSOR LOCATED ON PHASE L3													
	-BCT1	CURRENT TRANSFORMER LOCATED ON PHASE L1													
	-BCT2	CURRENT TRANSFORMER LOCATED ON PHASE L2													
	-BCT3	CURRENT TRANSFORMER LOCATED ON PHASE L3													
	-BCT4,-BCT7	CURRENT TRANSFORMER LOCATED ON PHASE L1													
	-BCT5,-BCT8	CURRENT TRANSFORMER LOCATED ON PHASE L2													
E	-BCT6,-BCT9	CURRENT TRANSFORMER LOCATED ON PHASE L3													E
	-BCZ	DISTANCE PROTECTION RELAY													
	-BEF	FREQUENCY PROTECTION RELAY													
	-BER	SUPERVISION RELAYS													
	-BES	SYNCHRONIZING RELAY													
	-BET	THERMAL PROTECTION RELAY													
	-BGA1	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CIRCUIT-BREAKER COMPARTMENT													
	-BGA2	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN BUSBAR COMPARTMENT (SINGLE BUSBAR SYSTEM) OR IN FRONT BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)													
	-BGA3	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CABLE COMPARTMENT													
	-BGA4	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN REAR BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)													
F	-BGB1...-BGB3	POSITION SWITCHES OF CIRCUIT-BREAKER													F
	-BGB4	POSITION SWITCH OF CIRCUIT-BREAKER: PASSING CONTACT CLOSING MOMENTARILY WHEN THE BREAKER OPENS													

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A	-BGI31	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS						A								
	-BGI32	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR							B							
	-BGI33	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR								C						
	-BGI41...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.									D					
	-BGI42	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT										E				
	-BGI43...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE											F			
	-BGI51	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT												A		
	-BGI52	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT													B	
	-BGI53	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT														C
	-BGK	POSITION SWITCH OF KEY LOCK						-EB3	HEATER LOCATED IN MOTOR CABINET														
-BGL1	POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB4	HEATER FOR VENTILATION TEST						E									
-BGS1...-BGS4	POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.							F								
-BGS5	PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE								A							
-BGT1	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE									B						
-BGT2	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1										C					
-BGT3	POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2											D				
-BGT4	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3												E			
-BGT5	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS													F		
-BGT6	POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1														A	
-BGT7	POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2															B
-BGT8	POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3						C									
-BGT11	PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER							D								
-BGT12	PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS								E							
-BGV1	POSITION SWITCH OF FRONT VENTILATOR						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1									F						
-BGV2	POSITION SWITCH OF REAR VENTILATOR						-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2										A					
-BM	HYGROSTAT						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3											B				
-BPA4	PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-FCT	THERMAL OVERLOAD RELEASES												C			
-BPA5	PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS													D		
-BPA6	PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR														E	
-BPA7	PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR															F
-BPS1	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER						-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK						A									
-BPS2	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2						-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE							B								
-BPS3	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3						-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE								C							
-BR	FLAME DETECTORS, SMOKE DETECTORS						-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS									D						
-BT	THERMOSTAT						-KFC	CLOSING RELAYS OR CONTACTORS										E					
-BT1	THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS						-KFI	INTEGRATED CIRCUITS											F				
-BT2	THERMOSTAT FOR ENERGIZATION OF VENTILATORS						-KFL	LOCKOUT RELAY												A			
-BT3	THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM						-KFM	MICROPROCESSORS													B		
-BU	BUCHHOLZ RELAY						-KFN	ANTIPUMPING RELAYS														C	
-BX1	UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING						-KFO	OPENING RELAYS OR CONTACTORS															D
-BX2	ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING						-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)						E									
-BZ,-BZ4	MULTIFUNCTION SENSORS						-KFR	RECLOSING RELAYS							F								
-BZ1	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1						-KFS	SYNCHRONIZING DEVICES								A							
-BZ2	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2						-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS									B						
-BZ3	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3						-KFZ	CONTROLLERS										C					
							-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER											D				
							-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD												E			
							-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1													F		
							-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2														A	
							-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS															B

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant				REFERENCE DESIGNATIONS MVD-33 20kV FlourMill SCHEMATIC DIAGRAM DRC BUS PT		Scale		= H05	F	
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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH	MANUAL CIRCUIT-BREAKERS					
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL	LINKS					
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM	MINIATURE SWITCH-DISCONNECTORS					
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS	SWITCH-DISCONNECTORS					
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT	TRUCK					
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING					
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE	EARTHING SWITCH					
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)					
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)					
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ	CLUTCHES					
B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA	ANTIFERRORESONANCE RESISTOR					
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD	DIODES					
	-PFB	BLUE SIGNAL LAMPS						-RAI	INDUCTORS					
	-PFF	FLAG RELAYS						-RAR	RESISTORS					
	-PFG	GREEN SIGNAL LAMPS						-RF	FILTERS					
C	-PFR	RED SIGNAL LAMPS						-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING					
	-PFS	SHORT CIRCUIT INDICATORS						-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT					
	-PFV	VOLTAGE INDICATORS						-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE					
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION					
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD					
	-PFW	WHITE SIGNAL LAMPS						-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1					
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2					
	-PFY	YELLOW SIGNAL LAMPS						-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2					
	-PGA	AMMETERS						-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS					
	-PGC	COUNTERS						-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4					
D	-PGF	FREQUENCYMETERS						-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5					
	-PGH	HOURMETERS						-SFA	AMMETRIC SWITCHES					
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS					
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL	LOCKING CONTACTS					
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO	OPENING PUSH-BUTTONS					
	-PGJ	ACTIVE ENERGY METERS						-SFR	RESET PUSH-BUTTONS					
	-PGK	REACTIVE ENERGY METERS						-SFS	SELECTOR SWITCHES					
	-PGM	MULTIFUNCTION INDICATORS						-SFT	TEST PUSH-BUTTONS					
	-PGP	POWER-FACTOR METERS						-SFU	UNLOCKING PUSH-BUTTONS					
	-PGQ	VARMETERS						-SFV	VOLTMETRIC SWITCHES					
E	-PGS	SYNCHRONOSCOPES						-TA	POWER TRANSFORMERS					
	-PGV	VOLTMETERS						-TB	CONVERTER					
	-PGW	WATTMETERS						-TB1	RECTIFIER FOR FIRST OPENING RELEASE					
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB2	RECTIFIER FOR SECOND OPENING RELEASE					
	-QAB	CIRCUIT-BREAKERS						-TB3	RECTIFIER FOR CLOSING RELEASE					
	-QAC	CONTACTORS (FOR POWER)						-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING					
	-QBD	DISCONNECTORS						-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT					
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE					
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER					
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)												
F	-QBD4	TWO-WAY DISCONNECTOR												
	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER												

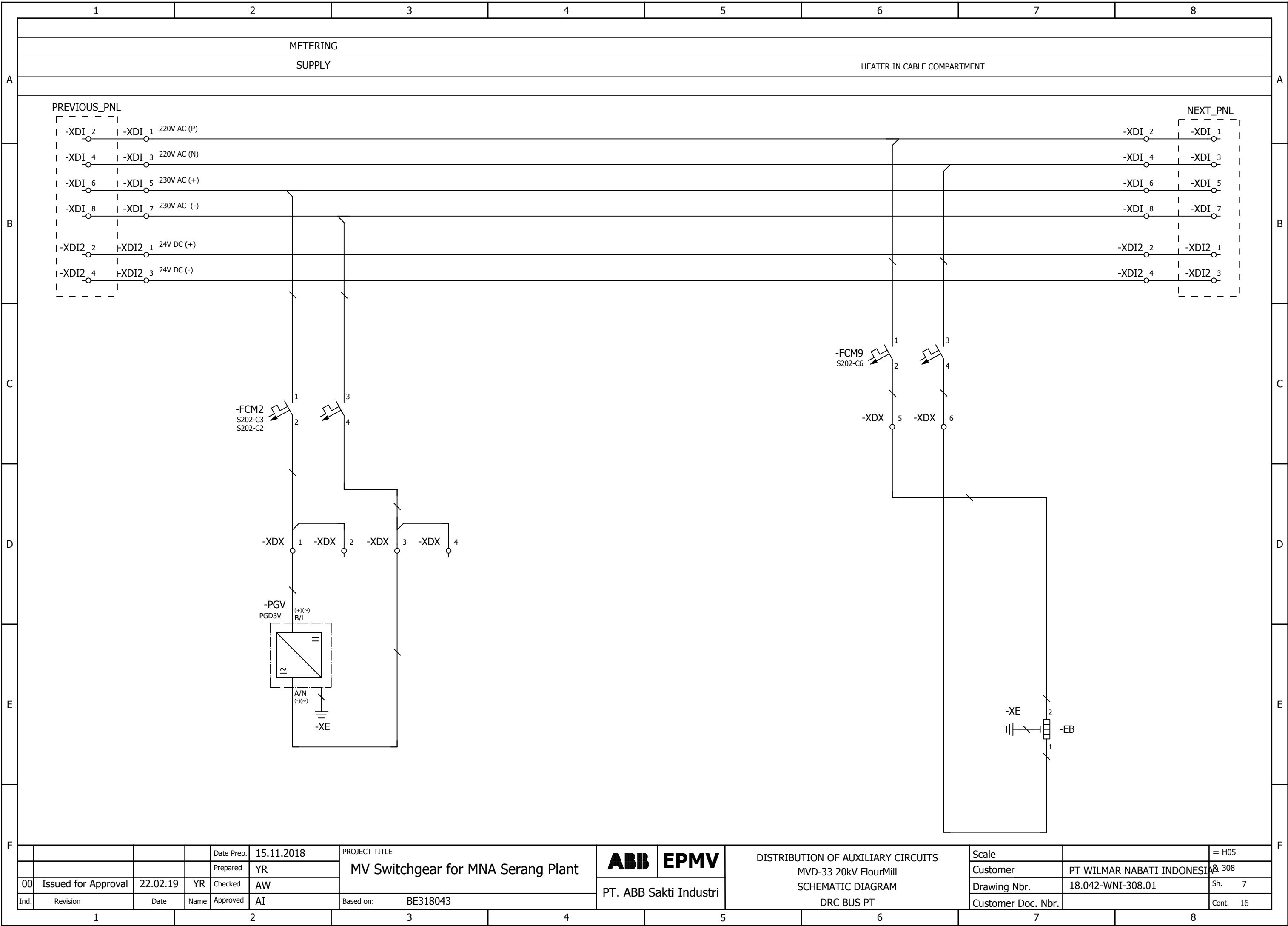
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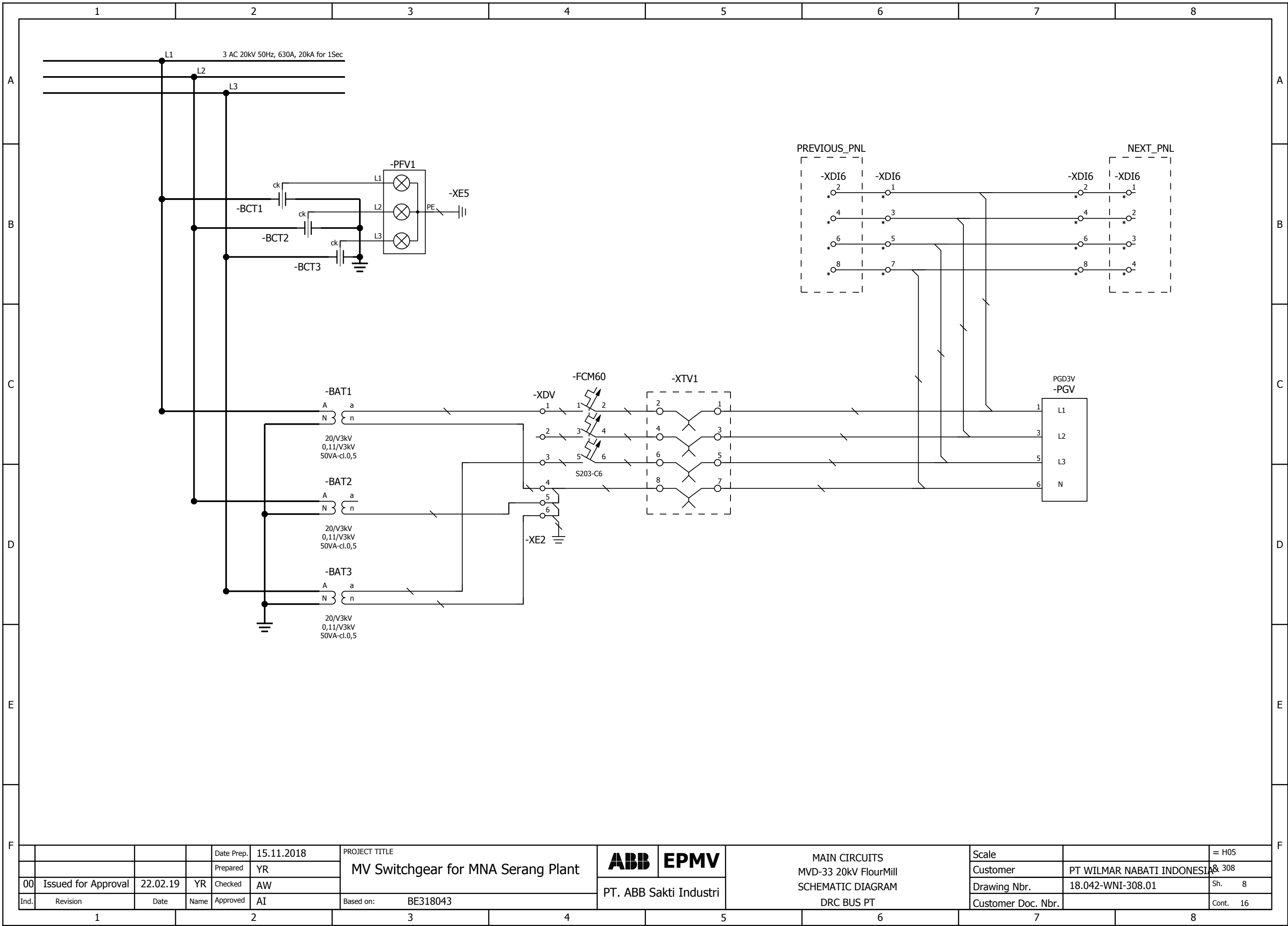
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A	-TFA	ACTIVE POWER TRANSDUCERS	-XD11	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS	A																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
	-TFC	CURRENT TRANSDUCERS	-XD12	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-TFF	FREQUENCY TRANSDUCERS	-XD13	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-TFJ	ACTIVE ENERGY TRANSDUCERS	-XD14	TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-TFK	REACTIVE ENERGY TRANSDUCERS	-XD16	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-TFM	MULTIFUNCTION TRANSDUCERS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
	-TFP	POWER-FACTOR TRANSDUCERS	-XD17	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-TFQ	REACTIVE POWER TRANSDUCERS	-XD18	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-TFS	SIGNAL CONVERTERS	-XD19	TERMINAL BLOCK FOR SPECIAL USE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-TFV	VOLTAGE TRANSDUCERS	-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
B	-WA	BUSBARS	-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS	B																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
	-WBC	POWER CABLES	-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-WE	EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.	-XDS	CURRENT SOCKETS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-WGC	CONTROL CABLES	-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-WGS	COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)	-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-WH	OPTICAL FIBRES	-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-XDA	TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS	-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-XDA1	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1	-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-XDA2	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2	-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-XDA3	CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3	-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
C	-XDA5	CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER	-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR	C																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
	-XDB	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS	-XDX	SUPPORT TERMINAL BLOCK																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)	-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER	-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-XDB10...-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS	-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS	-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR	-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR	-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR	-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
	-XDB95	CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
D	-XDC	CUSTOMER TERMINAL BLOCK			D																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
	-XDC1	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
	-XDC2	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
	-XDC3	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
	-XDC4	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
	-XDC5	CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
	-XDE	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
	-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
E	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS			E																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
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E

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490

580 mm

1

9.000

0.00

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0

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0

0.00

2

PTT

CUT OUT DIMENSION (in mm)

DIGITAL VOLTMETER

93

44

TEST BLOCK

5

80

14

17

94

				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	LV DOOR LAYOUT MVD-33 20kV FlourMill SCHEMATIC DIAGRAM DRC BUS PT	Scale		= H05
				Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT WILMAR NABATI INDONESIA & 308	
00	Issued for Approval	22.02.19	YR	Checked	AW			PT. ABB Sakti Industri	Drawing Nbr.		18.042-WNI-308.01	Sh.	9
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	Customer Doc. Nbr.			Cont.	16	

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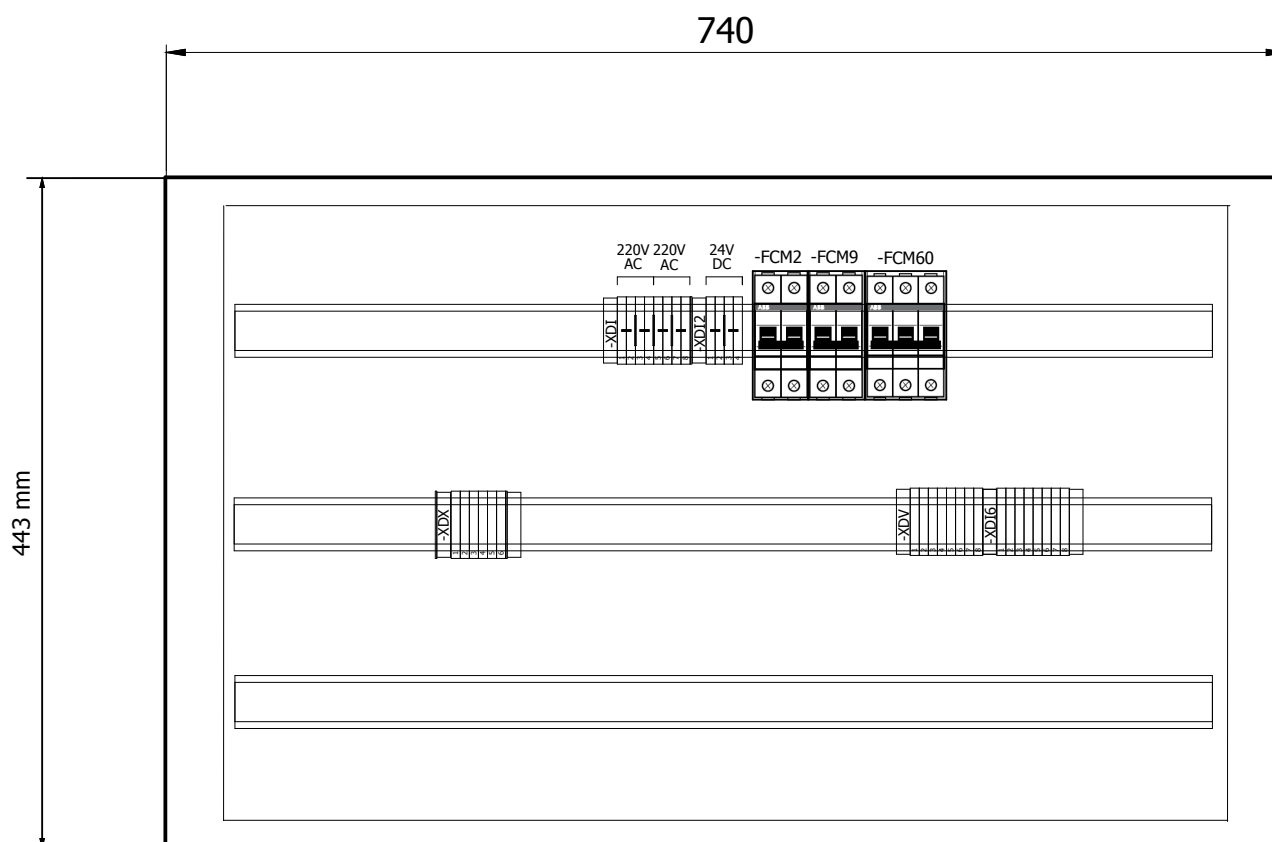
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	LV BOX LAYOUT MVD-33 20kV FlourMill SCHEMATIC DIAGRAM DRC BUS PT	Scale		= H05
				Prepared	YR					Customer	PT WILMAR NABATI INDONESIA	Sh. 308
00	Issued for Approval	22.02.19	YR	Checked	AW							
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318043	PT. ABB Sakti Industri		Drawing Nbr.	18.042-WNI-308.01	Sh. 10
										Customer Doc. Nbr.		Cont. 16

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F		E		D		C		B		A				
REMARK		CABLE	EQUIPMENT	Wire No	Terminal number	TYPE	WIRE NO.	EQUIPMENT	CABLE	REMARK				
==MVD-33=H05+-XDI6			-PREVIOUS_PNL-XDI6:2		1 ●	PT 2.5		-PGV:1		BLACK 1,5 mm²				
					2 ●	PT 2.5		-NEXT_PNL-XDI6:1						
				-PREVIOUS_PNL-XDI6:4		3 ●	PT 2.5		-PGV:3		BLACK 1,5 mm²			
					4 ●	PT 2.5		-NEXT_PNL-XDI6:2						
				-PREVIOUS_PNL-XDI6:6		5 ●	PT 2.5		-PGV:5		BLACK 1,5 mm²			
					6 ●	PT 2.5		-NEXT_PNL-XDI6:3						
				-PREVIOUS_PNL-XDI6:8		7 ●	PT 2.5		-PGV:6		BLACK 1,5 mm²			
					8 ●	PT 2.5		-NEXT_PNL-XDI6:4						
		REMARK: ☆ INTERPANEL LIGHT BUS,OFFERED TOGETHER WITH PANELS. △ OUTGOING CONTROL CABLE (Supply by Customer). ● BRIDGE BAR. ✕ PARTITION PLATE. (TEST JUMPER.												
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318043		PT. ABB Sakti Industri		Terminal diagram ==MVD-33=H05+-XDI6		Scale		= H05
					PROJECT TITLE		MV Switchgear for MNA Serang Plant		ABB EPMV		Customer		PT WILMAR NABATI INDONESIA	& 308
											Drawing Nbr.			Sh. 14
											Customer Doc. Nbr.			Cont. 16

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